

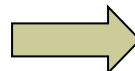
034IN - FONDAMENTI DI AUTOMATICA - FUNDAMENTALS OF AUTOMATIC CONTROL A.Y. 2024-2025

Part I: Motivations, Concepts, Examples

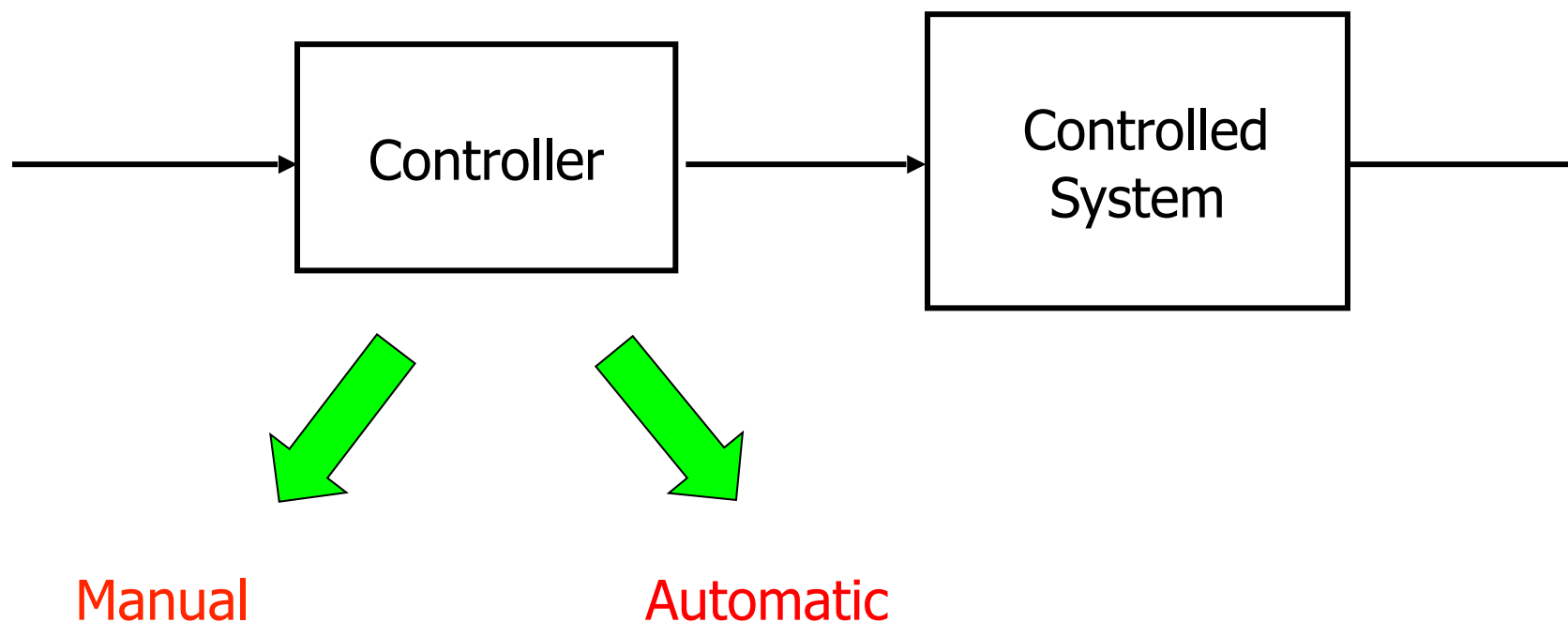
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Department of Engineering and Architecture

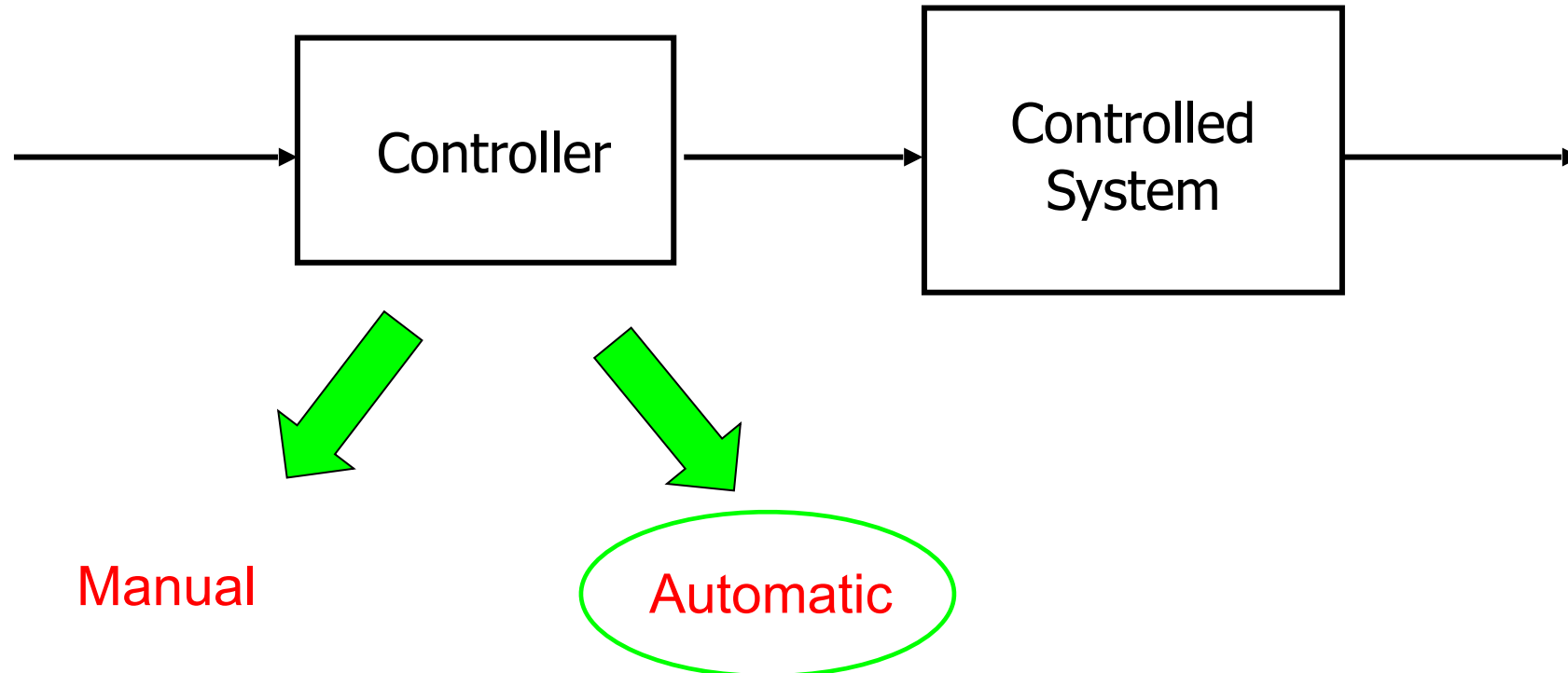
Content of the Module?



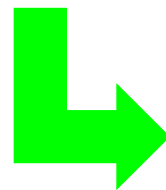
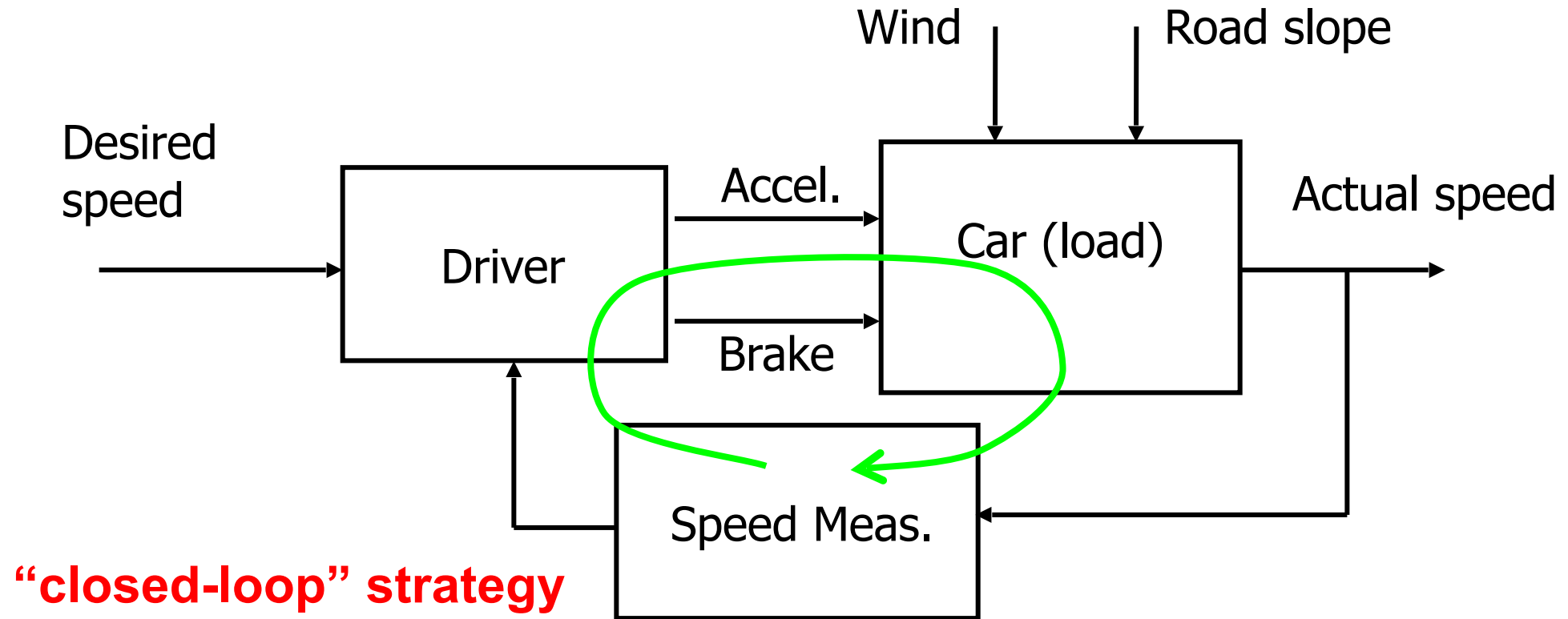
Methods and Tools to
analyse and design
automatic control systems
of practical relevance



To impose a given time-trajectory to one or more specific variables of an engineering system by acting on other variables that influence the behaviour of the system itself



Example: Speed Control



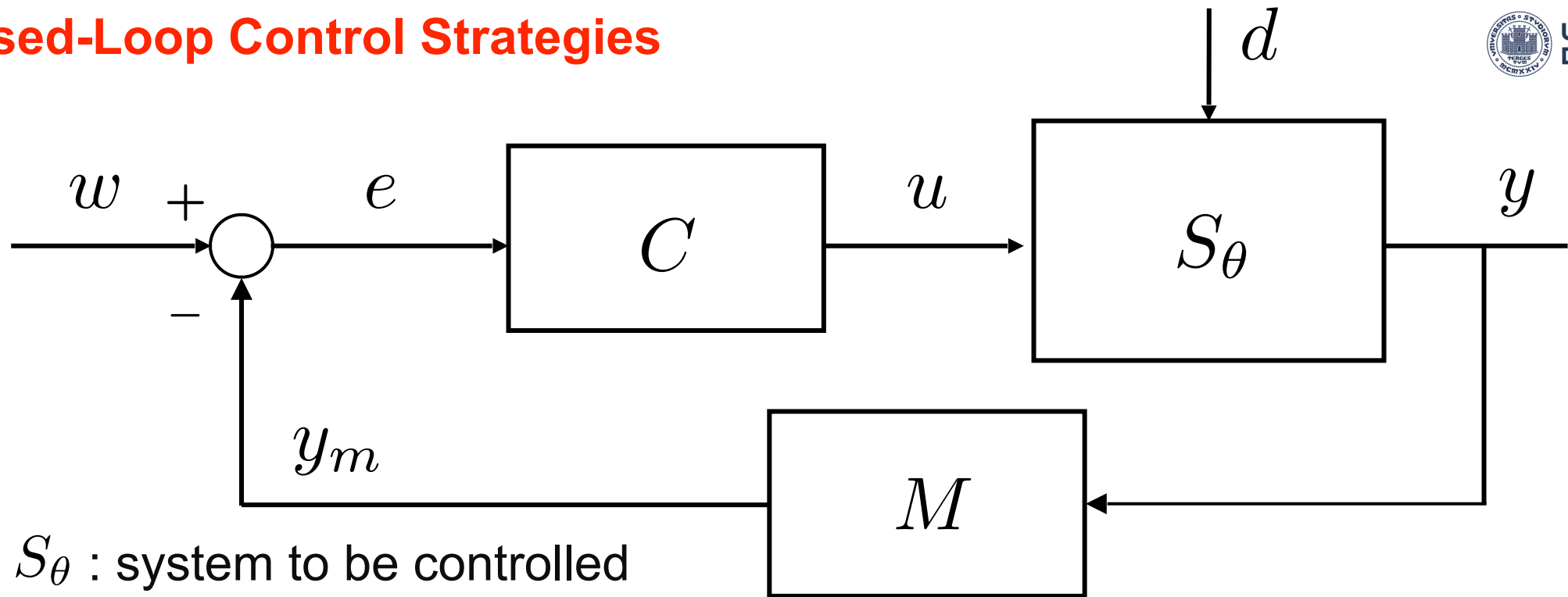
**Effective in the
presence of
uncertainty**



https://www.youtube.com/watch?v=IBC1nEq0_nk



Closed-Loop Control Strategies



- S_θ : system to be controlled
- C : controller
- θ : system's parameters
- y : controlled variable (output)
- u : control variable (accessible to the controller)
- d : disturbance
- w : reference variable (set-point)
- e : error
- y_m : measured output

The basic requirement is:

$$y(t) \simeq w(t)$$

Introducing the error variable $e(t) = w(t) - y(t)$, the above requirement becomes:

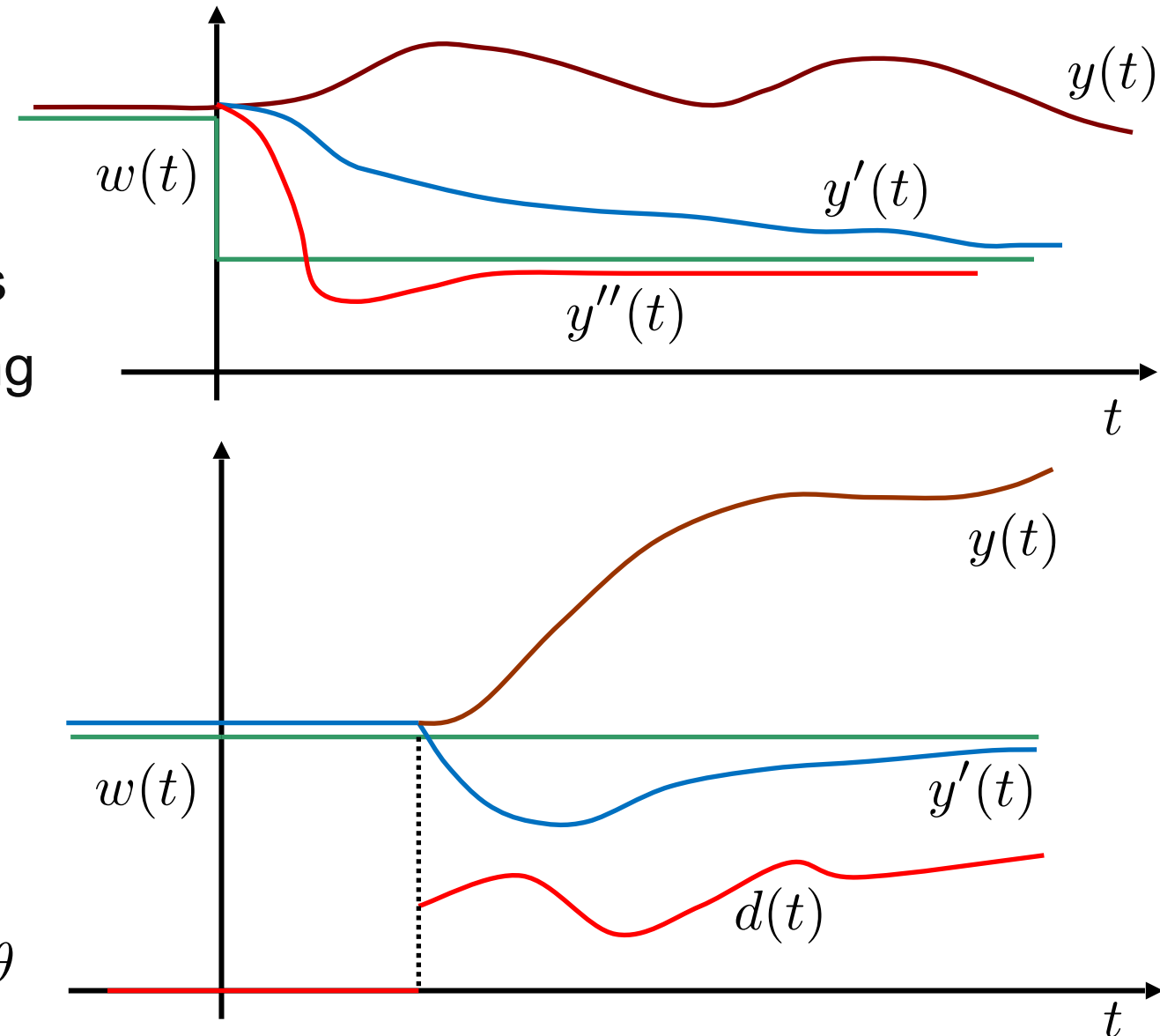
$$e(t) \simeq 0 \text{ in all situations "of interest"}$$

Without loss of generality, we focus on the continuous-time case. The discrete-time case is perfectly analogous

General Requirements of Control Systems

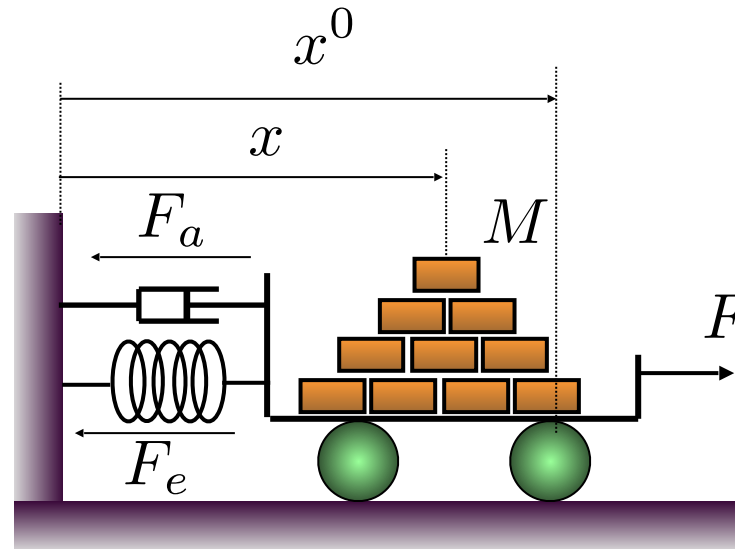


- (A) Static precision
 - ◆ In equilibrium conditions
- (B) Dynamic precision
 - ◆ Speed of response
 - ◆ Damping of possible oscillations
 - ◆ Capability of tracking fast-varying reference variables $w(t)$
- (C) Insensitivity to disturbances
 - ◆ Capability of rejecting a disturbance $d(t)$
- (D) Robustness
 - ◆ (A), (B), (C) in the presence of uncertain system's parameters θ



Example 2: Position Control

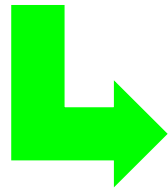
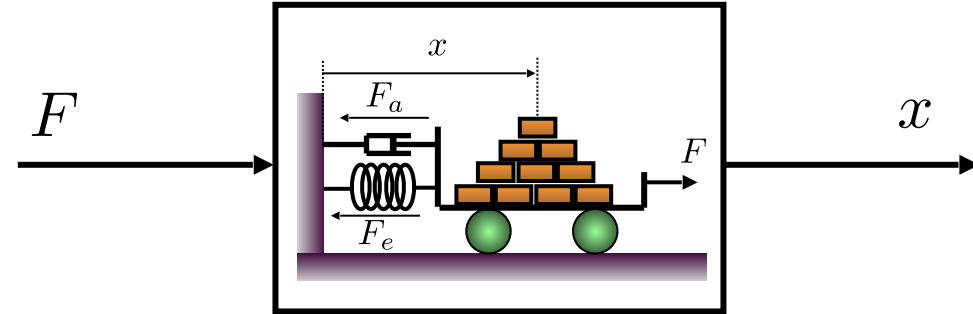
Mechanical system



- Control input: external force F
- Controlled output: position of the cart x
- Reference output: desired position $w = x^0$
- Elastic spring force: $F_e = kx$
- Damper viscous friction force: $F_a = h\dot{x}$

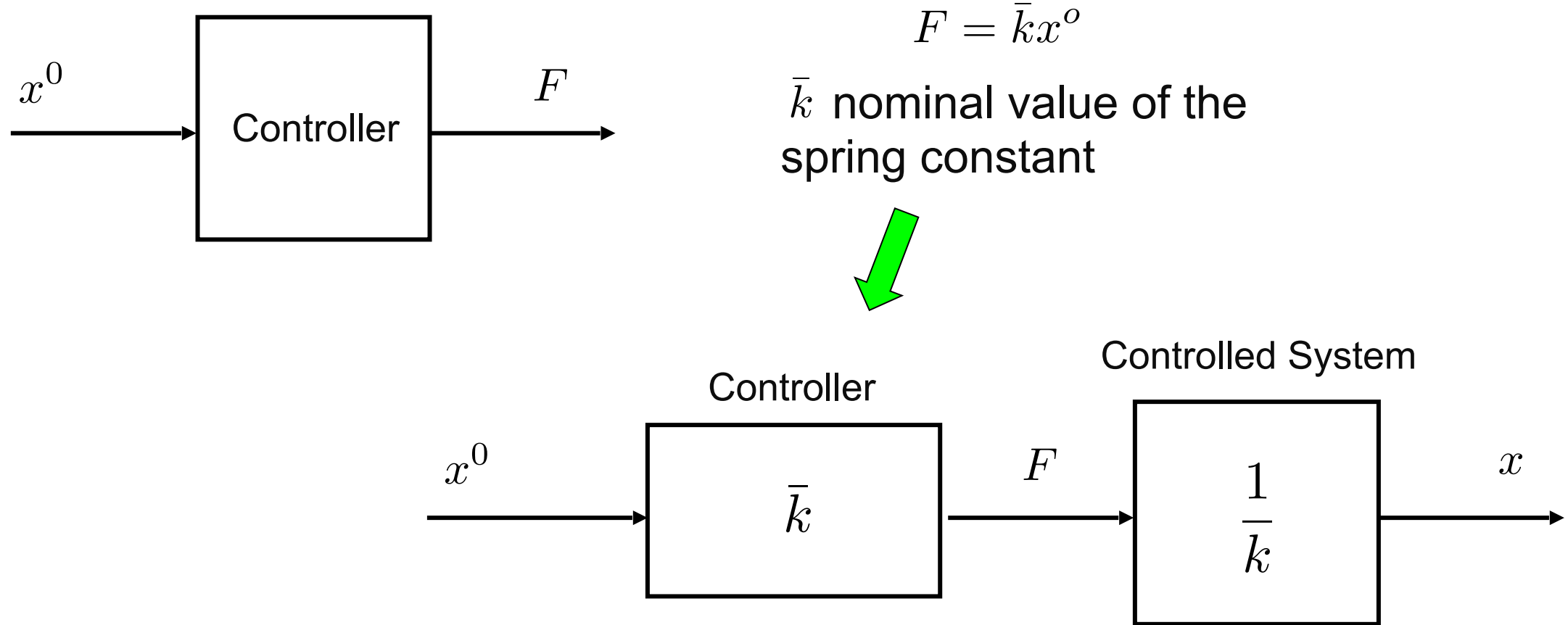
$$F = kx$$

Equilibrium of the forces
in **static** conditions



$$\text{output } x = \frac{1}{k} \text{input } F$$

Open-Loop Control Strategy

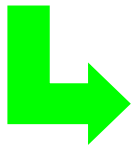


$$x = \frac{\bar{k}}{k}x^o \quad \longrightarrow \quad e = x^o - x = x^o \left(1 - \frac{\bar{k}}{k}\right)$$

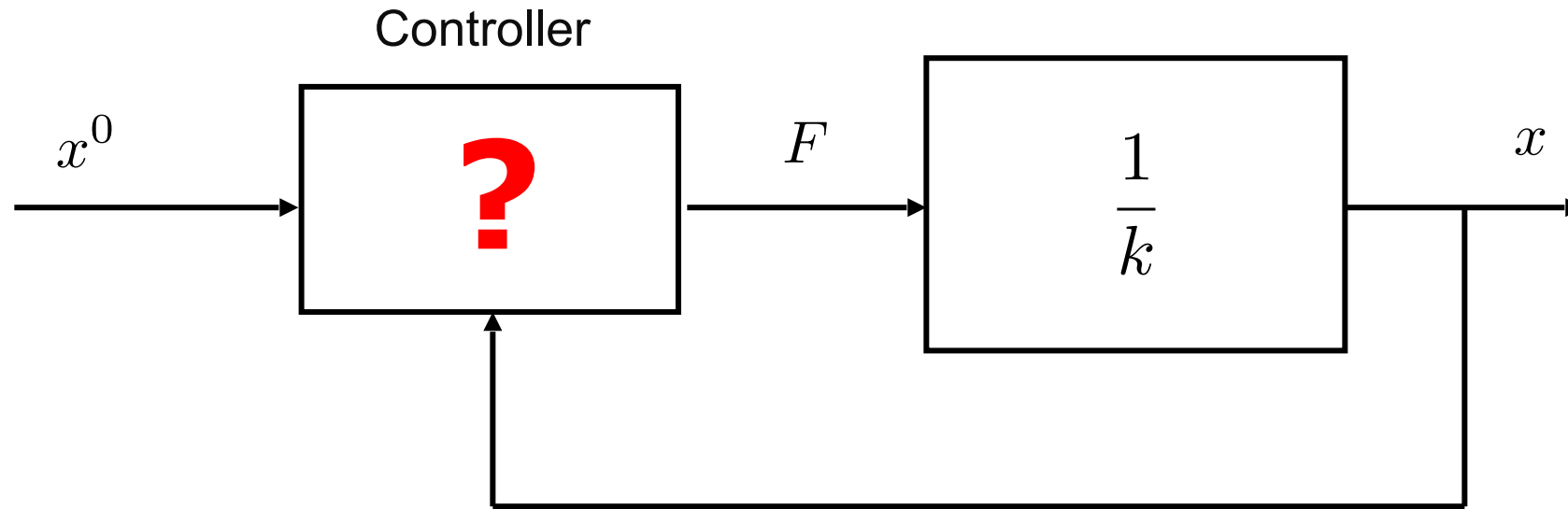
- In nominal conditions $k = \bar{k} \longrightarrow e = 0$
- In uncertain conditions $k \neq \bar{k} \longrightarrow e = x^o \frac{\Delta k}{k} \neq 0$

$$\Delta k = k - \bar{k} \neq 0$$

uncertainty



There is no way to compensate for the uncertainty by an open-loop control strategy



Let us opt for a **proportional control** strategy:

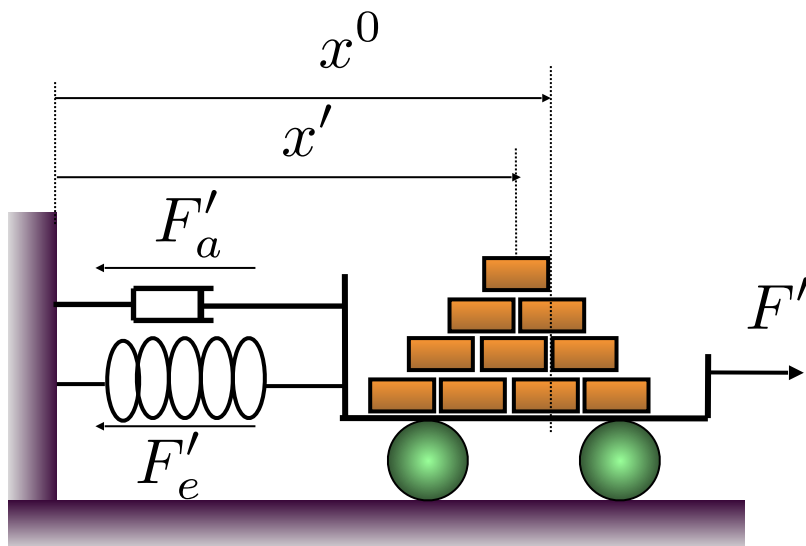
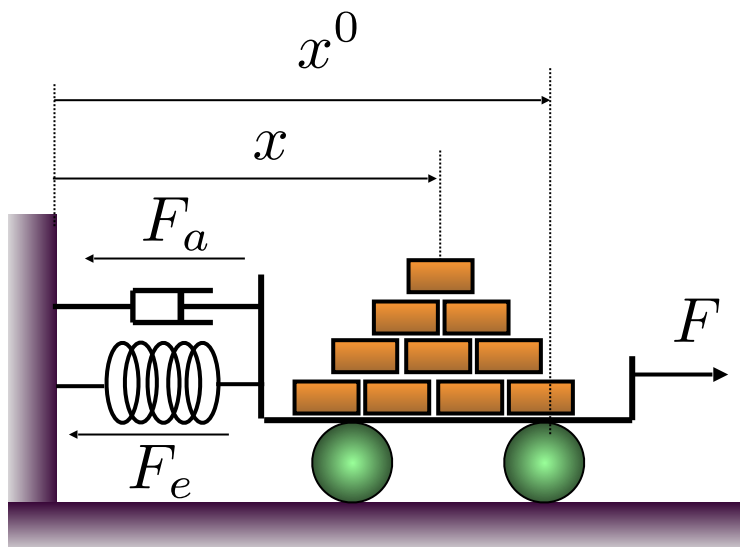
$$F = \alpha \underbrace{(x^o - x)}_e, \quad \alpha > 0$$

Suppose $x^o \neq 0$:

$$x = \frac{1}{k}F = \frac{1}{k}\alpha (x^o - x) \longrightarrow x \left(1 + \frac{\alpha}{k}\right) = \frac{\alpha}{k}x^o$$

$$\longrightarrow x = \frac{\alpha/k}{1 + \alpha/k} x^o \longrightarrow e = x^o - x = \frac{1}{1 + \alpha/k} x^o$$

- In nominal conditions $k = \bar{k} \longrightarrow e \neq 0$
- In uncertain conditions $k \neq \bar{k} \longrightarrow e \simeq 0$ if $\alpha \gg k_{\max}$



$$F = \alpha \underbrace{(x^0 - x)}_e, \quad \alpha > 0$$

The effects of mass, spring, and damper are not negligible, and dynamic behaviours such as oscillations may occur

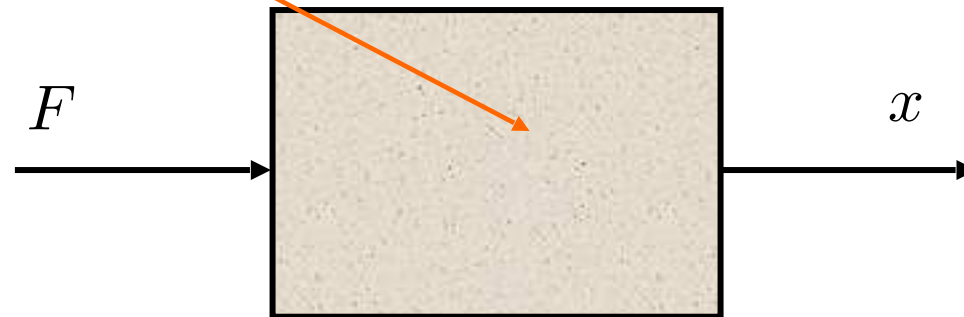


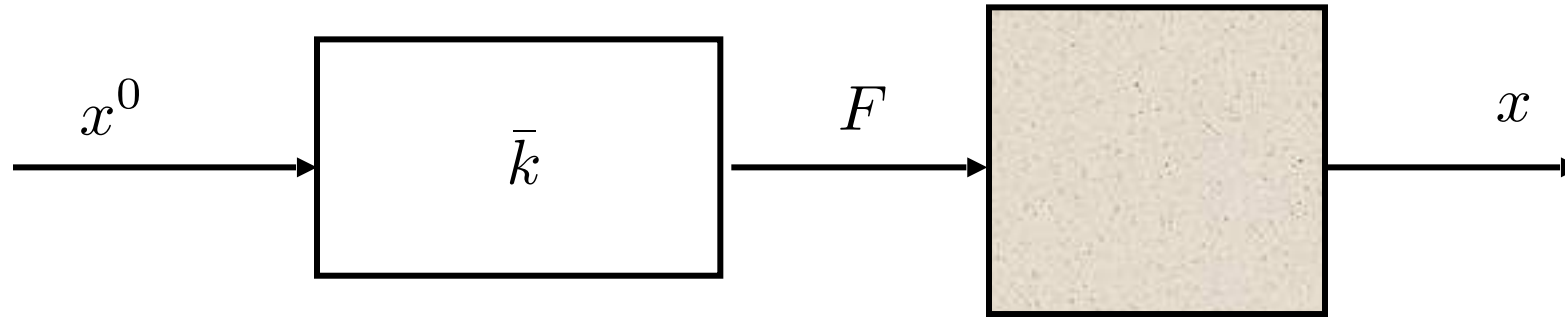
Models capturing the dynamic modes of behaviour are clearly necessary

sum of all external forces = $M\ddot{x}$


$$F - kx - h\dot{x} = M\ddot{x}$$


$$M\ddot{x} + h\dot{x} + kx = F$$





$$\underbrace{M\ddot{x} + h\dot{x} + kx}_{\text{dynamic terms}} = \underbrace{\bar{k}x^o}_{\text{constant}}$$

condition for static equilibrium

$$\underbrace{\begin{matrix} x(0) \\ \dot{x}(0) \end{matrix}}_{\text{initial conditions}}$$



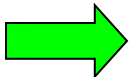
[Livescripts in MS Teams](#): see Part 1:
[CART_OPENLOOP_CONTROL](#)



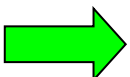
From differential equations theory:

$$M\ddot{x} + h\dot{x} + kx = \bar{k}x^o$$

← constant term due to open-loop control

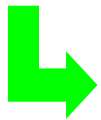
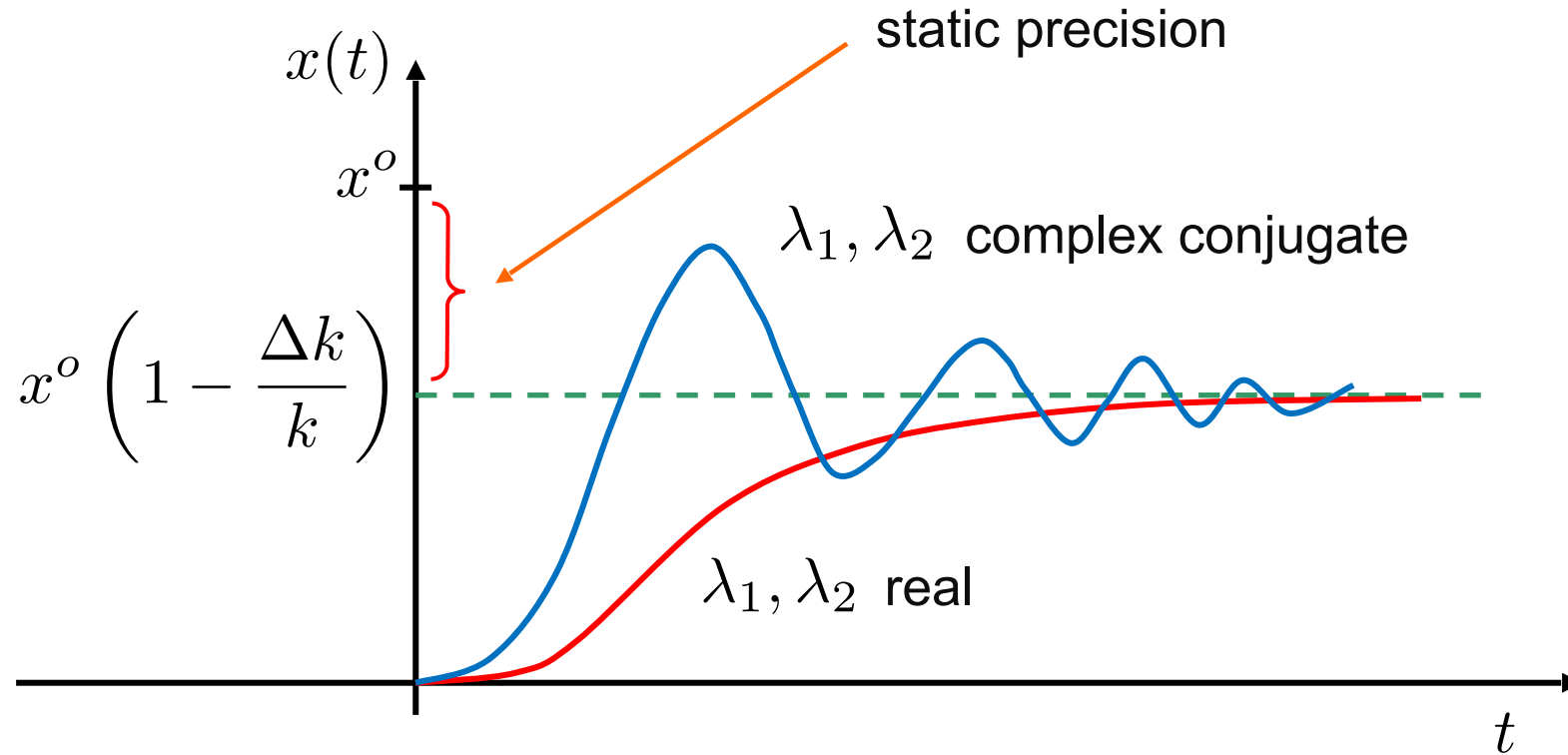
 $M\lambda^2 + h\lambda + k = 0$  λ_1, λ_2

 polynomial algebraic equation  roots

- If λ_1, λ_2 are real  $x(t) = c_1 e^{\lambda_1 t} + c_2 e^{\lambda_2 t} + c_3$

- If λ_1, λ_2 are complex conjugate, that is $\lambda_1 = \sigma + j\omega$, $\lambda_2 = \sigma - j\omega$

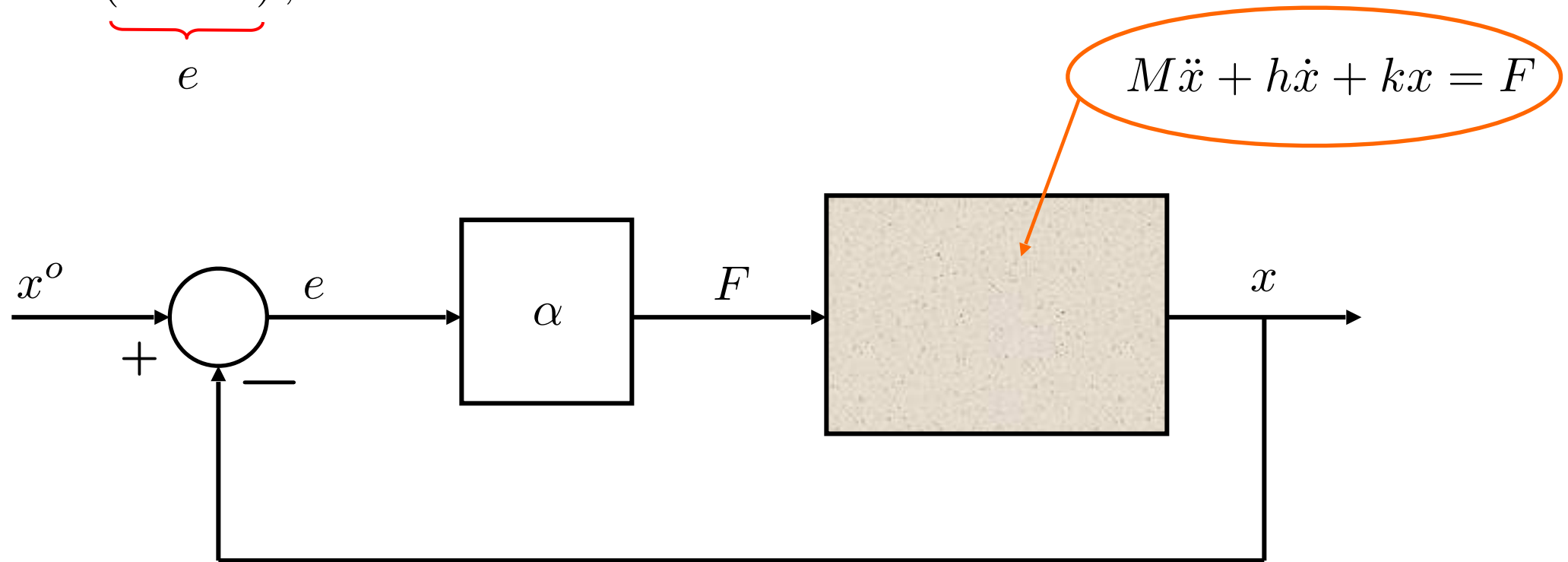
 $x(t) = c_4 e^{\sigma t} \cos(\omega t + c_5) + c_6$



By the open-loop control strategy the dynamic behaviour cannot be modified since it only depends on the system's parameters M, k, h and not on the controller. Hence dynamic precision cannot be modified by the controller

Let us use again a **proportional closed-loop control** strategy:

$$F = \alpha \underbrace{(x^o - x)}_e, \quad \alpha > 0$$



Plugging in the proportional control scheme we get:

[Livescripts in MS Teams](#): see Part 1:
[CART_CLOSEDLOOP_P_CONTROLLER](#)



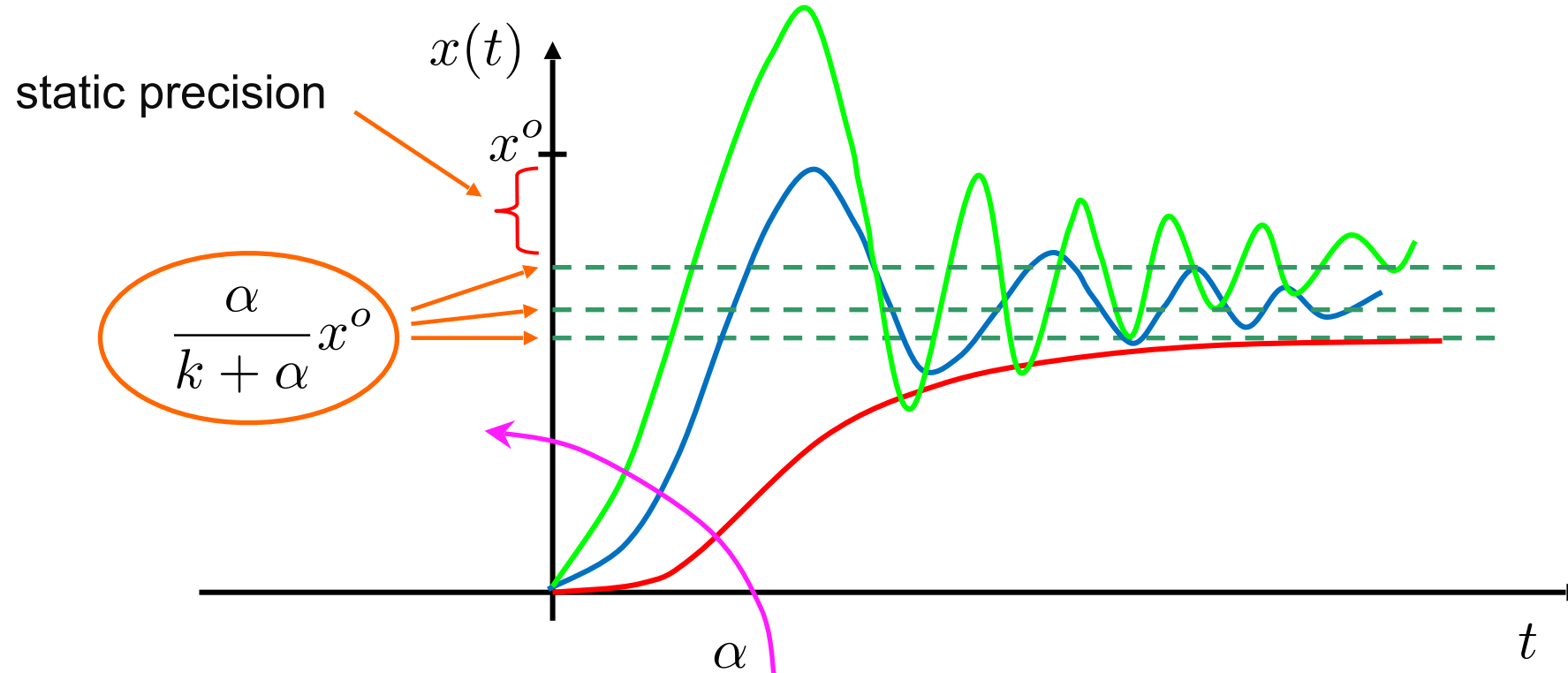
$$M\ddot{x} + h\dot{x} + kx = \alpha(x^o - x)$$

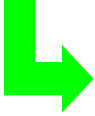
↳ $M\ddot{x} + h\dot{x} + (k + \alpha)x = \alpha x^o$

the constant term of the algebraic equation is **influenced by the controller gain α**

↳ $M\lambda^2 + h\lambda + \underbrace{(k + \alpha)}_{\text{polynomial algebraic equation}} = 0 \quad \Rightarrow \quad \underbrace{\lambda_1, \lambda_2}_{\text{roots}}$

↳ **The roots λ_1, λ_2 can be modified by choosing α !!**



- 
- The dynamic precision also depends on the choice of the control gain
 - Static and dynamic requirements are in contrast with each other: better static precision implies worse dynamic precision

A Different Closed-Loop Control Strategy



Let us now opt for a **proportional/derivative** control scheme:



$$F = \alpha (x^o - x) + \beta \frac{d}{dt} (x^o - x), \quad \alpha, \beta > 0$$

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`CART_CLOSEDLOOP_PD_CONTROLLER`




$$M\ddot{x} + h\dot{x} + kx = \alpha(x^o - x) - \beta\dot{x}$$


$$M\ddot{x} + (h + \beta)\dot{x} + (k + \alpha)x = \alpha x^o$$

the first-order term of the algebraic equation is influenced by the **derivative controller gain** β


$$M\lambda^2 + (h + \beta)\lambda + (k + \alpha) = 0$$

polynomial algebraic equation


$$\lambda_1, \lambda_2$$

roots

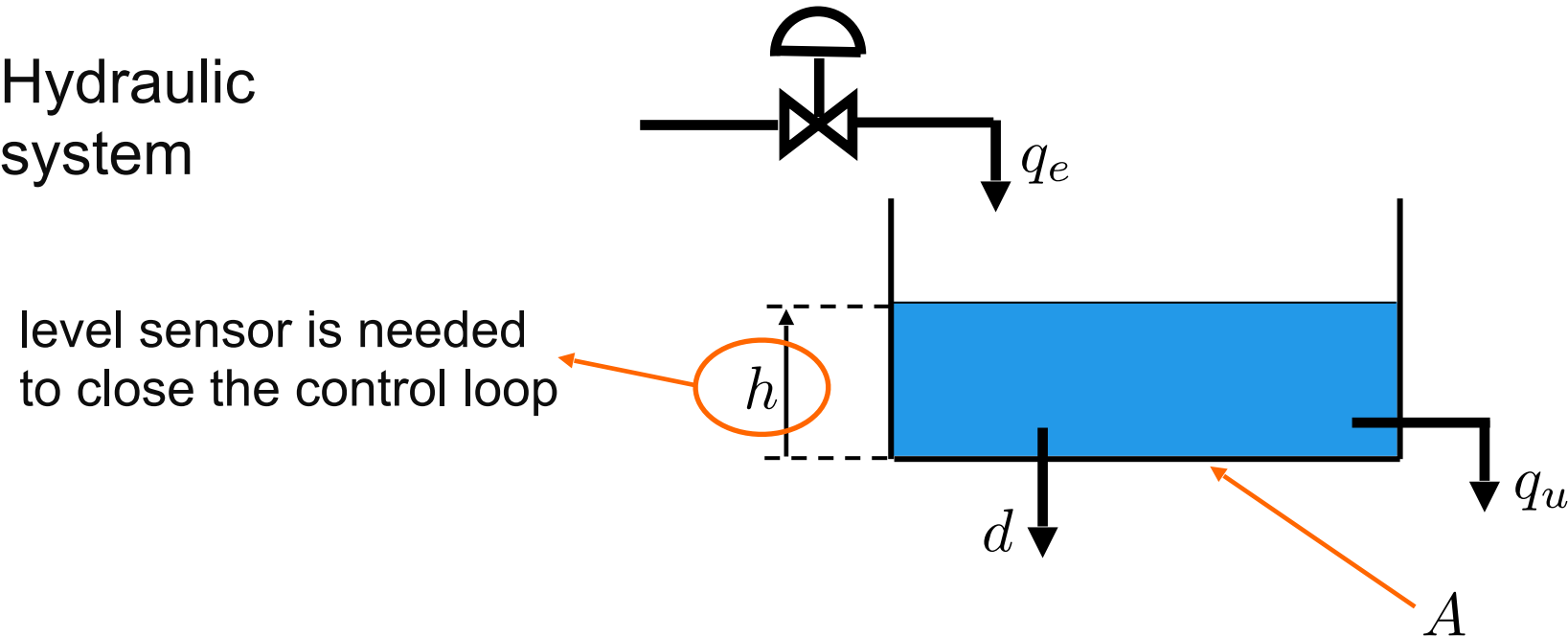


The roots λ_1, λ_2 can be modified by choosing α, β !!

the constant term of the algebraic equation is influenced by the **proportional controller gain** α

Example 3: Tank Level Closed-Loop Control

Hydraulic
system



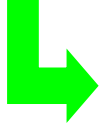
- Control input: input flow-rate q_e
- Controlled output: level of liquid in the tank h
- Reference output: constant desired level in the tank $w = h^o$
- Out-flow rate: $q_u = kh$
- Disturbance: d

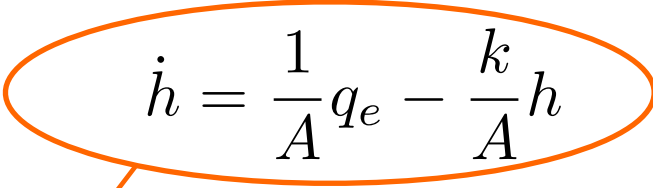
By the usual assumption of infinite-height of the tank and supposing (for the moment) that $d = 0$:

$$A\dot{h} = q_e - kh$$

conservation equation:

volume time-variation = input flow-rate - out flow-rate

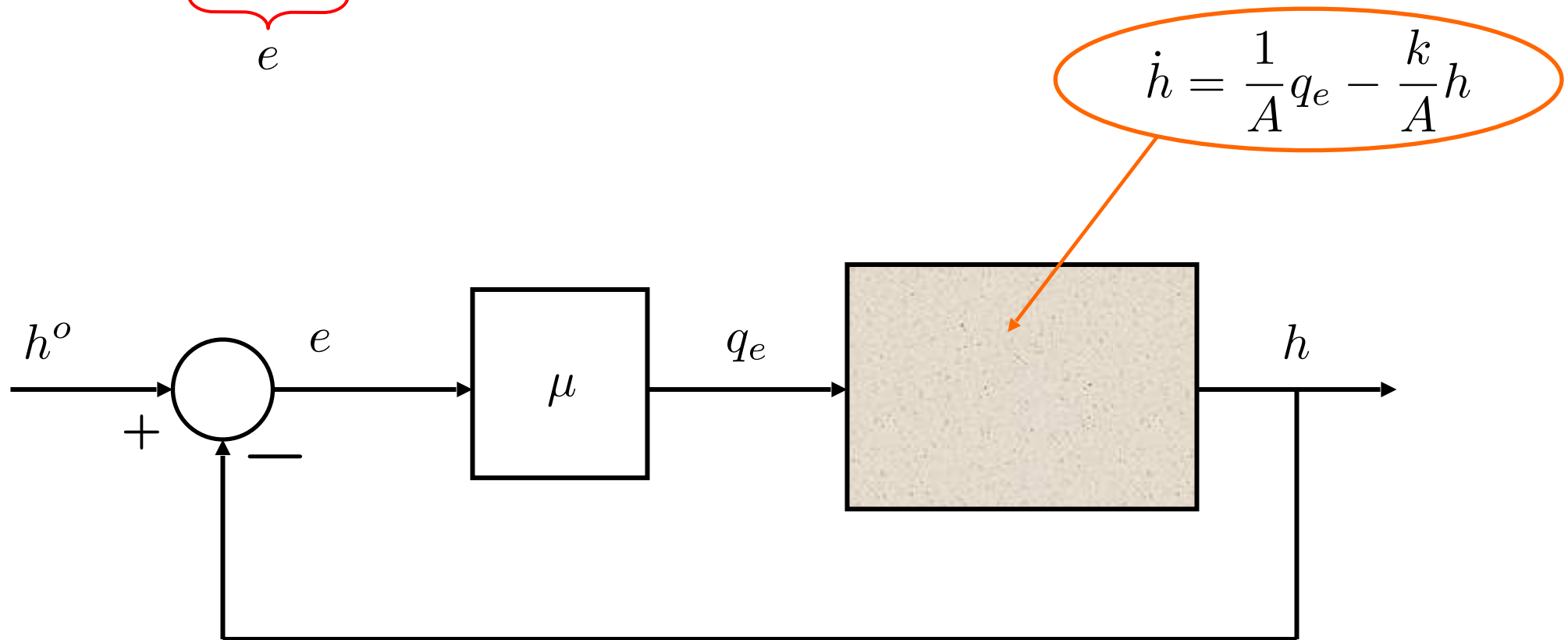

$$\dot{h} = \frac{1}{A}q_e - \frac{k}{A}h$$


$$\dot{h} = \frac{1}{A}q_e - \frac{k}{A}h$$



Let us opt for a **proportional control** strategy:

$$q_e = \mu \underbrace{(h^o - h)}_e, \quad \mu > 0$$



Plugging-in the proportional control scheme we get:

$$\dot{h} = \frac{1}{A} \mu (h^o - h) - \frac{k}{A} h$$


$$\dot{h} = - \underbrace{\frac{1}{A} (k + \mu)}_{\sigma > 0} h + \underbrace{\frac{\mu}{A} h^o}_{\text{constant}}$$

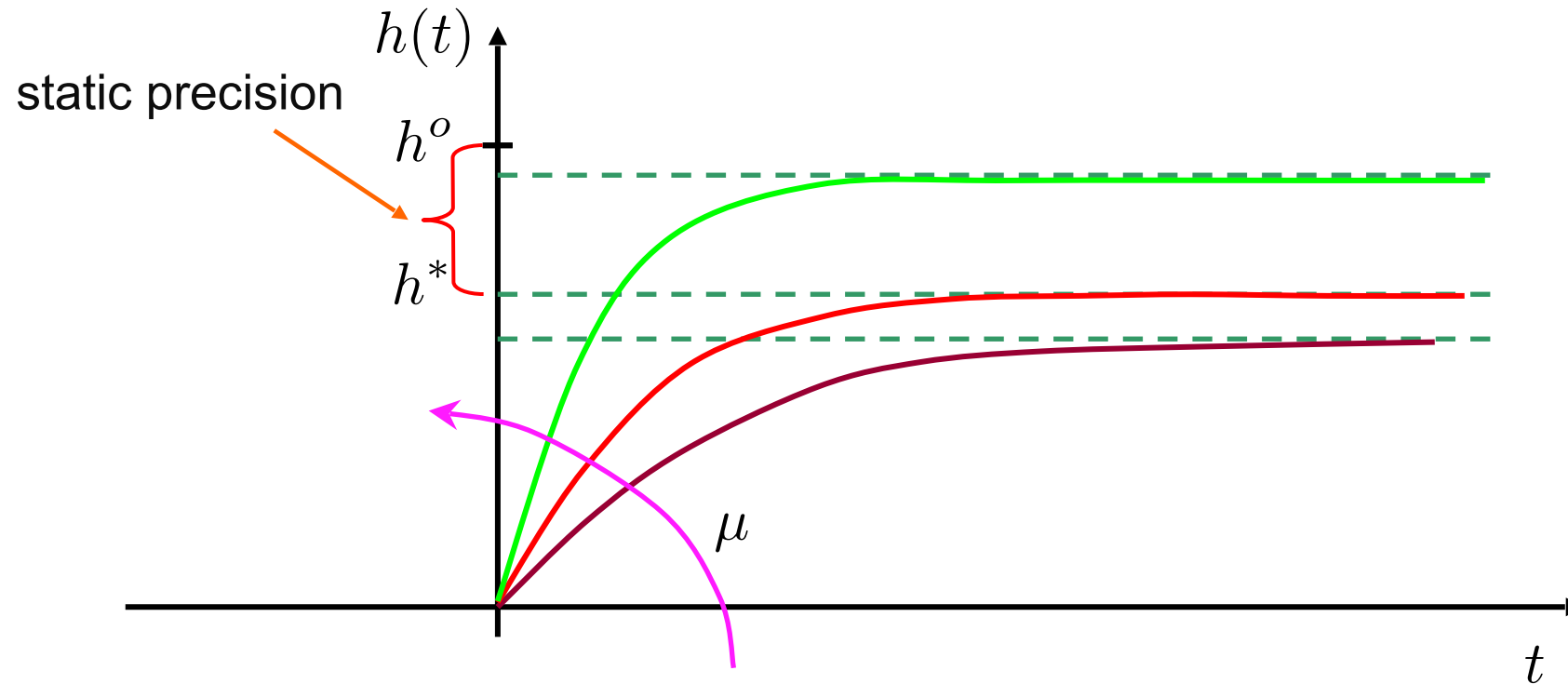
Assuming $h(0) = 0$ we get:

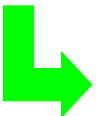
$$h(t) = \underbrace{\frac{\mu}{\mu + k}}_{h^*} h^o (1 - e^{-\sigma t}), t \geq 0$$

$$h^* = \lim_{t \rightarrow \infty} h(t)$$

[Livescripts in MS Teams](#): see Part 1:
TANK_CLOSEDLOOP_P_CONTROLLER





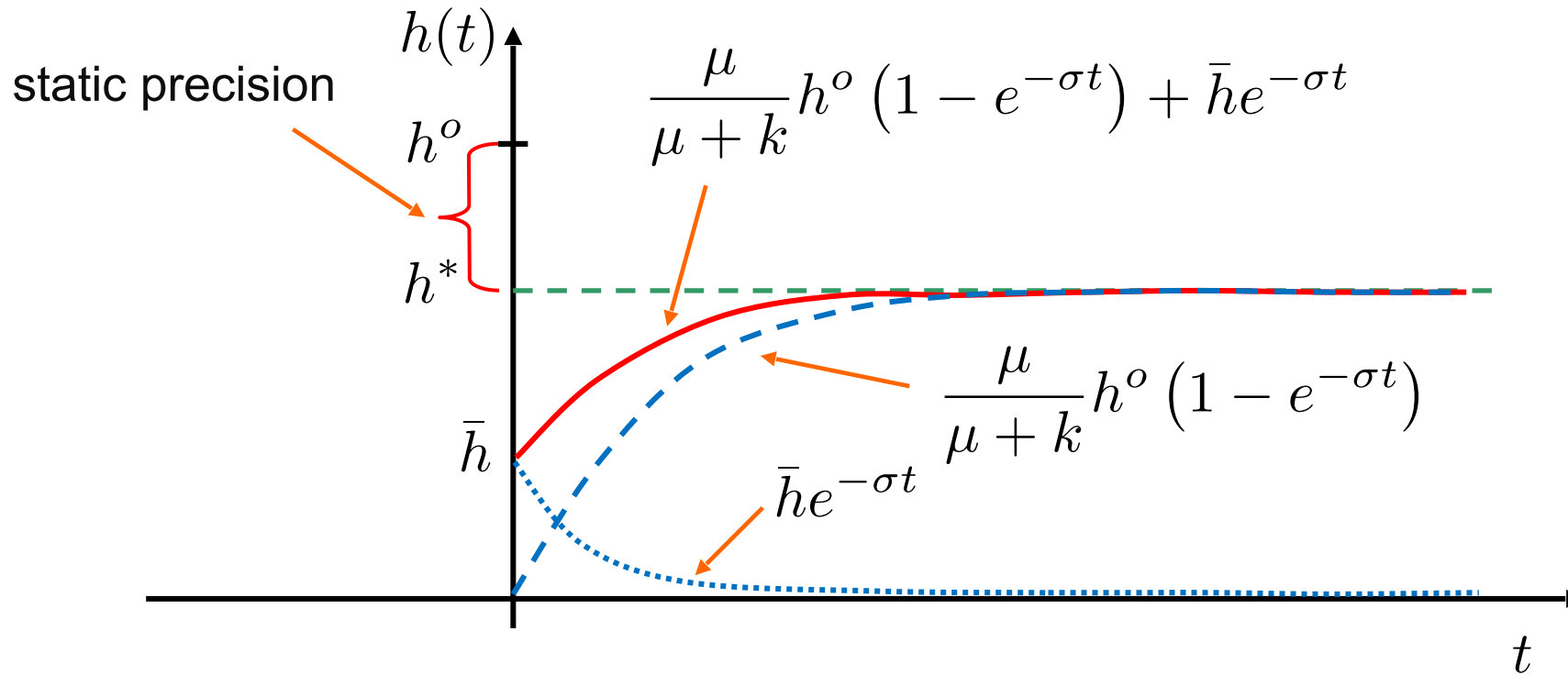
- 
- By increasing the control gain μ both the static and the dynamic precision improve
 - In this hydraulic example, static and dynamic requirements are not in contrast with each other

Assuming $h(0) = \bar{h} > 0$ we get:

effect of the non-zero initial condition

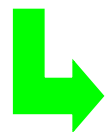
$$h(t) = \underbrace{\frac{\mu}{\mu + k} h^o (1 - e^{-\sigma t})}_{\text{effect of the control input}} + \overbrace{\bar{h} e^{-\sigma t}}^{\text{effect of the non-zero initial condition}}, t \geq 0$$

effect of the control input

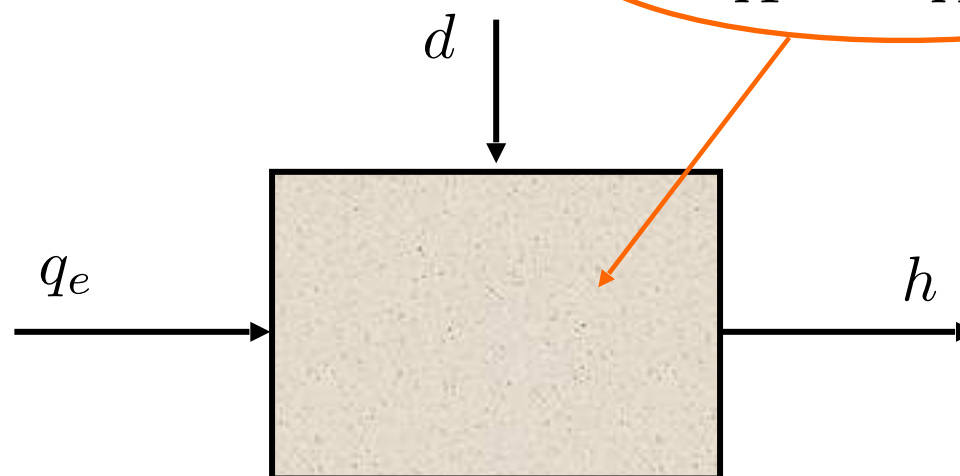
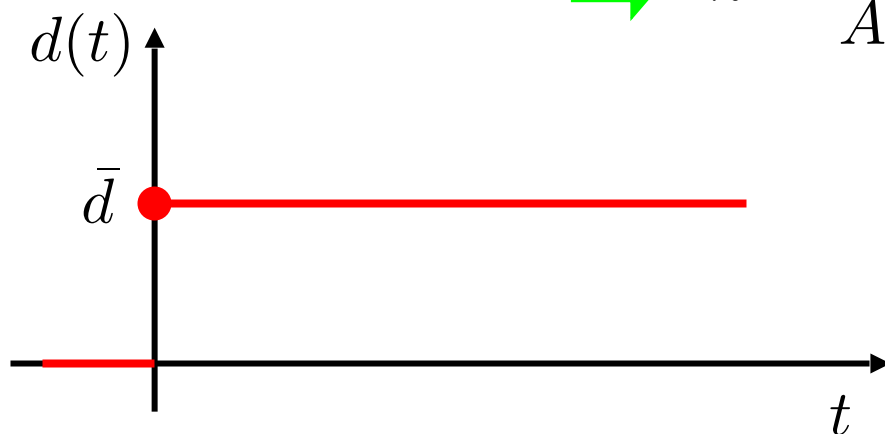


By the usual assumption of infinite-height of the tank and supposing that a step-like disturbance $d(t) = \bar{d}$, $t \geq 0$ (leakage) is acting on the tank:

$$A\dot{h} = q_e - kh - \bar{d}$$



$$\dot{h} = \frac{1}{A}q_e - \frac{k}{A}h - \frac{1}{A}\bar{d}$$



$$\dot{h} = \frac{1}{A}q_e - \frac{k}{A}h - \frac{1}{A}\bar{d}$$

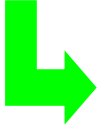
Plugging in the proportional control scheme we get:

$$q_e = \mu (h^o - h), \quad \mu > 0$$


$$\dot{h} = \frac{1}{A} \mu (h^o - h) - \frac{k}{A} h - \frac{1}{A} \bar{d}$$

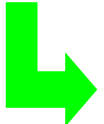

$$\dot{h} = - \underbrace{\frac{1}{A} (k + \mu)}_{\sigma > 0} h + \underbrace{\frac{\mu}{A} h^o - \frac{1}{A} \bar{d}}_{\text{constant}}$$

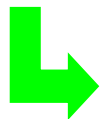
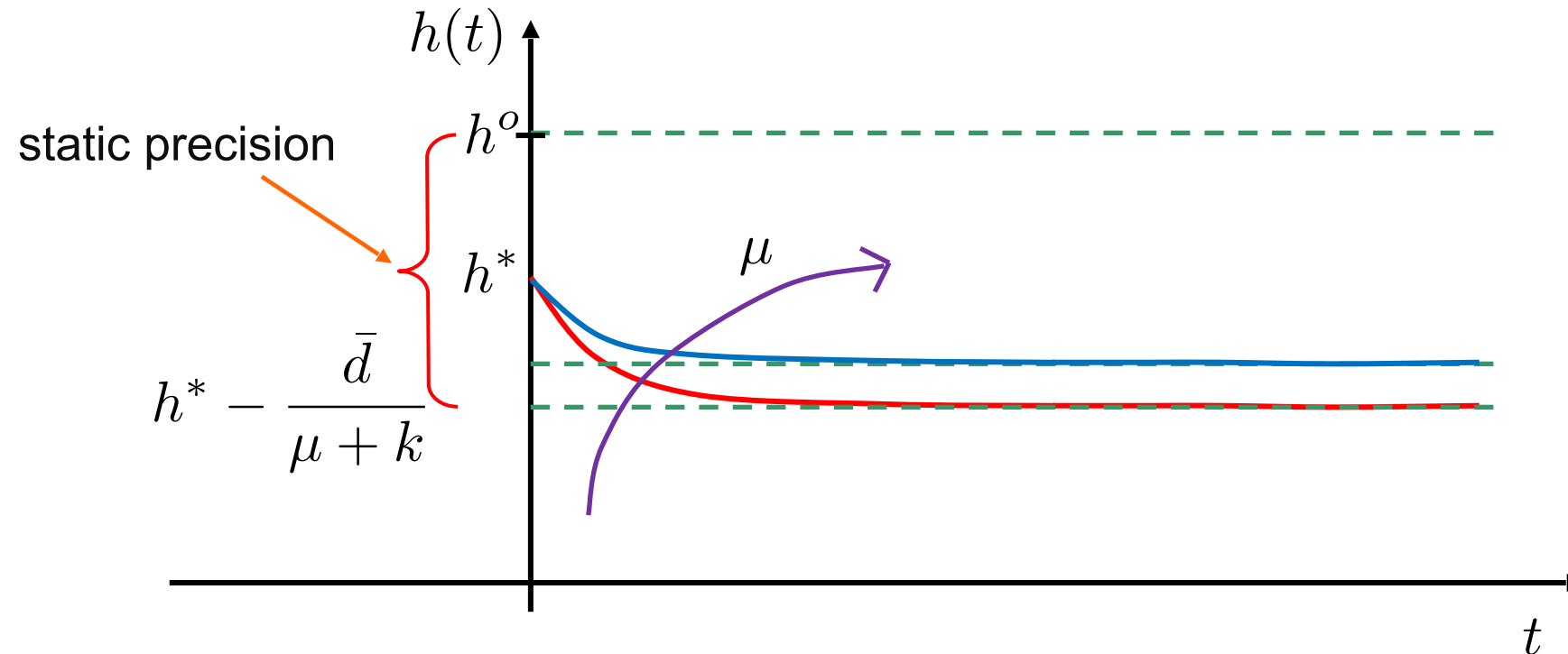
For simplicity suppose: $h(0) = h^* = \frac{\mu}{\mu + k} h^o$


$$h(t) = \frac{\mu}{\mu + k} h^o (1 - e^{-\sigma t}) + h(0)e^{-\sigma t} - \frac{\bar{d}}{\mu + k} (1 - e^{-\sigma t})$$

$$= h^* - \cancel{h^* e^{-\sigma t}} + \cancel{h^* e^{-\sigma t}} - \frac{\bar{d}}{\mu + k} (1 - e^{-\sigma t})$$

$$h(t) = h^* - \frac{\bar{d}}{\mu + k} (1 - e^{-\sigma t}), t \geq 0$$


$$\lim_{t \rightarrow \infty} h(t) = h^* - \frac{\bar{d}}{\mu + k}$$



Increasing the proportional gain μ improves the disturbance rejection performance

How to Improve the Static Precision?

A) Adding open-loop compensation actions:

open-loop action

$$q_e = \mu (h^o - h) + \overbrace{kh^o + \bar{d}}$$


$$\dot{h} = \frac{1}{A}\mu (h^o - h) + \frac{k}{A}h^o + \cancel{\frac{1}{A}\bar{d}} - \frac{k}{A}h - \cancel{\frac{1}{A}\bar{d}}$$


$$h(t) = h^o (1 - e^{-\sigma t}), t \geq 0$$
$$\lim_{t \rightarrow \infty} h(t) = h^o$$


But:

to implement the open-loop compensation actions, the knowledge of $k, \bar{d}$ is required



How to Improve the Static Precision?



B) Modifying the closed-loop controller by adding an **integral action** thus obtaining a **proportional/integral (PI) controller**:

[Livescripts in MS Teams](#): see Part 1:
`TANK_CLOSEDLOOP_PI_CONTROLLER`



$$q_e = \mu [h^o - h(t)] + \varphi \int_0^t [h^o - h(\tau)] d\tau \quad \mu, \varphi > 0$$

Justification:

$$q_e = \mu e(t) + \varphi \int_0^t e(\tau) d\tau \quad \longrightarrow \quad \dot{q}_e = \mu \dot{e}(t) + \varphi e(t)$$

Equilibrium:

$$\begin{cases} e(t) = \text{cost} = \bar{e} \\ q_e(t) = \text{cost} \end{cases} \quad \longrightarrow \quad 0 = \mu \cdot 0 + \varphi \cdot \bar{e}$$

$$\bar{e} = 0$$

